Statistical Power

The Power Formula

$$\text{power} \;=\; \Phi\left(\frac{|\mu_t - \mu_c|\sqrt{N}}{2\sigma} \;-\; \Phi^{-1}\left(1 - \frac{\alpha}{2}\right)\right)$$

Read it as a subtraction: **what you expect** − **what you need** = how much room you have. Feed that difference into the table.

$$\underbrace{\frac{|\mu_t - \mu_c|\sqrt{N}}{2\sigma}}_{\text{what you expect, in standard errors}} \qquad \underbrace{\Phi^{-1}\left(1 - \tfrac{\alpha}{2}\right)}_{\text{the bar you must clear}} \qquad \underbrace{\text{their difference}}_{\text{look this up in the table}}$$

The Minimum Detectable Effect (MDE)

$$\text{MDE} \;=\; \left(\Phi^{-1}\left(1 - \tfrac{\alpha}{2}\right) + \Phi^{-1}(\text{target power})\right) \;\times\; \frac{2\sigma}{\sqrt{N}}$$

The same equation, solved for the effect instead of for power. At $\alpha = 0.05$ and 80% target power the leading bracket is $1.96 + 0.84 = \mathbf{2.80}$.

Notation

Symbol	Name	What it is
N	Sample size	Total subjects collected, across both arms.
μ_t	Treatment mean	Average outcome under treatment.
μ_c	Control mean	Average outcome under control.
$\mu_t - \mu_c$	Effect size	The true difference between the two means: the *signal*. Abbreviated δ in code.
σ	Standard deviation	How much outcomes vary from one another: the *noise*.
α	Significance level	Type I error rate you tolerate — claiming an effect when there is none. Sets the bar.
β	Type II error rate	Probability of missing an effect that is real.
power	$1 - \beta$	Probability of detecting an effect that is real.
Φ	Normal CDF	Area to the *left* of a z-score. This is what the table gives you.
Φ^{-1}	Inverse normal CDF	Its inverse: hand it an area, it returns the z-score.

Critical Values Worth Memorising

Quantity	Value	Where it shows up
$\Phi^{-1}(0.975)$	**1.96**	The bar, for a two-sided test at $\alpha = 0.05$
$\Phi^{-1}(0.95)$	1.645	The bar, for a one-sided test at $\alpha = 0.05$
$\Phi^{-1}(0.80)$	**0.84**	Target power of 80%, inside the MDE formula
$\Phi^{-1}(0.90)$	1.282	Target power of 90%

Your Work

Reading the Table, Both Directions

	Forwards: $z \to$ area (Φ)	**Backwards:** area $\to z$ (Φ^{-1})
You have	A z-score, e.g. 0.28	An area, e.g. 0.975
You do	Split it: **0.2** picks the row, **.08** picks the column	Hunt the area in the *body* of the table
You read	The cell where they meet	The row and column *margins* of that cell
You get	$\Phi(0.28) = .6103$, i.e. **power** $= 0.61$	Row 1.9 + column .06, i.e. $z = 1.96$

The table below gives $\Phi(z)$ — the area to the **left** of z — for $z \geq 0$. For negative z, use $\Phi(-z) = 1 - \Phi(z)$.

Standard Normal Table: $\Phi(z)$

z	**.00**	**.01**	**.02**	**.03**	**.04**	**.05**	**.06**	**.07**	**.08**	**.09**
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	**.6103**	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	**.7995**	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	**.8508**	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	**.9750**	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

The shaded cells are the four lookups used in the session: $\Phi(0.28) = .6103$, $\Phi(0.84) = .7995$, $\Phi(1.04) = .8508$, and $\Phi(1.96) = .9750$.